

Sec 7.1 and 7.2 – Integration Review

Basic Antiderivatives

Test yourself! Complete each of these antiderivative rules to see what you remember. Practice these rules so that they are automatic so you can recognize when to use them and focus your attention on the harder processes of integrals.

$$\text{For } n \neq -1, \int x^n dx = \frac{x^{n+1}}{n+1} \quad \int \frac{1}{\cos^2(x)} dx = \underline{\tan(x)}$$

$$\int \frac{1}{x} dx = \underline{\ln(x)} \quad \int e^x dx = \underline{e^x}$$

$$\int \sin(x) dx = \underline{-\cos(x)} \quad \int \frac{1}{1+x^2} dx = \underline{\arctan(x)}$$

$$\int \cos(x) dx = \underline{\sin(x)} \quad \int \frac{1}{\sqrt{1-x^2}} dx = \underline{\arcsin(x)}$$

Using Algebra and Substitution to Evaluate Integrals

There are many methods of integration and we are only getting to a few in Calculus II! In all methods, you should think of algebra as a tool to let you make changes to the function you are integrating to make your work easier, as illustrated by these examples which only use the methods of antiderivatives and substitution.

$$\int (3x+3)(x+1) dx = \int \underline{x^2 + 4x + 3} dx = \underline{\frac{1}{3}x^3 + 2x^2 + 3x} + C$$

algebra: multiply

$$\int \frac{4-3t^4+t^8}{t} dt = \int \left(\frac{4}{t} - 3t^3 + t^7 \right) dt = 4 \ln|t| - \frac{3}{4}t^4 + \frac{t^8}{8} + C$$

algebra: divide

$$\int \frac{\sin(x)}{\cos(x)+3} dx = \int \frac{-dw}{w} = \int \frac{-1}{w} dw = -\ln|w| + C = -\ln|\cos(x)+3| + C$$

substitution:

$$w = \cos(x) + 3$$

$$dw = -\sin(x) dx$$

$$-dw = \sin(x) dx$$

$$\int \frac{x^2+2}{1+x^2} dx = \int \frac{x^2}{1+x^2} dx + \int \frac{2}{1+x^2} dx = \int \frac{\frac{1}{2}w}{w} dw + \int \frac{2}{1+x^2} dx$$

$$= \frac{1}{2} \ln|w| + 2 \arctan(x) + C$$

$$= \frac{1}{2} \ln|1+x^2| + 2 \arctan(x) + C$$

.. $\frac{1}{2} dw = 2x dx$

$$\int (x^3 - 1)^4 x^2 dx$$

Solution: Use **substitution**, where $w = x^3 - 1$. This is the whole polynomial inside parentheses and has a derivative of $3x^2$, which is a constant multiple of the rest of the term.

Therefore, choosing $w = x^3 - 1$, $dw = 3x^2 dx$, so $\frac{1}{3} dw = x^2 dx$. Using this substitution,

$$\int (x^3 - 1)^4 x^2 dx = \int \frac{1}{3} w^4 dw = \frac{1}{15} w^5 + C = \frac{1}{15} (x^3 - 1)^5 + C.$$

$$\int \frac{4}{x^2} - \frac{3}{x^2 + 1} dx$$

Solution: Use **antiderivatives** directly: each term is a constant multiple of a standard derivative, since $\frac{4}{x^2} = 4x^{-2}$.

$$\int \frac{4}{x^2} - \frac{3}{x^2 + 1} dx = -4x^{-1} - 4 \arctan(x) + C$$

$$\int \frac{t^3 - 3t^2 + 1}{t^2} dt$$

Solution: First, we will use algebra to simplify and then we will be able to use standard antiderivatives.

$$\int \frac{t^3 - 3t^2 + 1}{t^2} dt = \int t - 3 + t^{-2} dt = \frac{1}{2} t^2 - 3t - t^{-1} + C$$

$$\frac{d}{dx} \int_2^x t^3 e^t dt$$

Solution: This problem is asking us to take the derivative of an integral. We know these operations essentially undo each other and using the FTC, we can write

$$\frac{d}{dx} \int_2^x t^3 e^t dt = x^3 e^x.$$

If we wanted to think through the process, we could consider $F(t)$ as the antiderivative of $t^3 e^t$, which means $\frac{d}{dx} F(x) = x^3 e^x$. Then

$$\frac{d}{dx} \int_2^x t^3 e^t dt = \frac{d}{dx} [F(x) - F(2)] = \frac{d}{dx} F(x) - \frac{d}{dx} F(2) = x^3 e^x - 0.$$

$$\frac{d}{dw} \int_{-2}^{\sin(w)} e^{-x^2} dx$$

Solution: Since we have the derivative of a definite integral - with the variable of w that we are differentiating by part of the limits of integration - we can use the FTC. Since the limit is a function of w , we use the chain rule:

$$\frac{d}{dw} \int_{-2}^{\sin(w)} e^{-x^2} dx = e^{-\sin^2(w)} \cos(w).$$

To understand where the chain rule comes from, we could consider $F(x)$ as the antiderivative of e^{-x^2} , which means $\frac{d}{dz} F(z) = e^{-z^2}$. Then

$$\frac{d}{dw} \int_{-2}^{\sin(w)} e^{-x^2} dx = \frac{d}{dw} [F(\sin(w)) - F(-2)] = \frac{d}{dw} F(\sin(w)) - \frac{d}{dw} F(-2) = F'(\sin(w)) \cos(w) - 0 = e^{-\sin^2(w)} \cos(w)$$

$$\int x \cos x \, dx = x \cdot \sin x - \int \sin x \, dx = x \cdot \sin x + \cos x + C$$

$$u = x \quad v = \sin x$$

$$u' = 1 \quad v' = \cos x$$

$$\int \cos^2 x \, dx = \int \cos x \cdot \cos x \, dx = \sin x \cdot \cos x + \int \sin^2 x \, dx$$

$$u = \cos x \quad v = \sin x \qquad \sin^2 x + \cos^2 x = 1$$

$$u' = -\sin x \quad v' = \cos x \qquad \sin^2 x = 1 - \cos^2 x$$

$$\int \cos^2 x \, dx = \sin x \cdot \cos x + \int 1 - \cos^2 x \, dx$$

$$\int \cos^2 x \, dx = \sin x \cdot \cos x + \int 1 \, dx - \int \cos^2 x \, dx$$

$$2 \int \cos^2 x \, dx = \sin x \cdot \cos x + x$$

$$\int \cos^2 x \, dx = \frac{1}{2} (\sin x \cdot \cos x + x) + C$$

$\int \ln(2 + x) \, dx$ Integration by parts doesn't really lead anywhere at first, so try substitution

$w = 2 + x = \int \ln(w) \, dw$ and **now** try integration by parts

$$dw = dx \quad u = \ln w \quad u' = \frac{1}{w} \quad v = w \quad v' = 1 \rightarrow$$

$$\int \ln(w) \, dw = w \cdot \ln w - \int \frac{1}{w} w \, dw = w \cdot \ln w - w$$

$$= (2 + x) \cdot \ln(2 + x) - (2 + x) + C$$